Lecture 6

Observational Data: Stratifying and Modelling

(Please, sign in on iClicker)

Today's Learning Objectives

1. Define a set of variables that control for confounding.
2. Analyze stratified observational data.
3. Interpret regression models fittings for observational data.
4. Supply appropriate caveats for causal claims from fitting regression models to observational data.
5. Comment on the differing goals of regression for prediction versus regression for explanation.

Outline

1. Overview of Observational Studies
2. The Western Collaborative Group Study Data
3. Key Statistical Concepts
4. Stratified Analysis for Causality

Where are we at?

- We have been devoting most of our time to randomized experiments (i.e., A/B or A/B/n testing).
- This is the ideal setting where we actually have control over the data collection and treatment randomization.
- Moreover, this class of statistical framework allows us to draw causal conclusions on a given outcome of interest.

Review

- Recall that if we can randomize experimental units to the two groups of conditions being compared (e.g, control treatment A and experimental treatment B), then data analysis and causal interpretation can be clear.
- Hence, we can be confident in our A/B testing decisions. Furthermore, this treatment randomization setting will be crucial to control for confounding variables on a high level.

For example

- As long as we have an adequate overall sample size (via Power Analysis!) and randomize properly, we can gather statistical evidence to state that website version B directly causes better sales than website version A.
- Consequently, we will be convinced that a permanent switch to website version B will be profitable.
- Moreover, suppose all website visitors could hypothetically be labelled as either "fussy" or "easy-to-please."

Therefore

- It is tremendously unlikely to end up with a much higher proportion of fussy people in the website version A than in B via treatment randomization.
- Consequently, excuses like "A did not win because it was faced with a harder-to-please crowd" will not stand up.

1. Overview of Observational Studies

We will shift gears to observational studies.
Unlike randomized experiments, we can only gather data from a population of interest via a proper and representative sample.
In observational studies, the researcher WILL NOT have control over the variable of interest X.

1.1. Without randomization, life is much harder!

For instance, if people self-select whether to see the A or B website version, you would really want to measure their fussiness level and use this when analyzing the data.

This practice will allow you to estimate accurate and precise effect estimates of your variable X on the response Y.

Observational Study Strategy

In general, the observational study strategy includes:

1. To the extent possible, recording potential confounding variables as part of the data collection process.
2. Using these potential confounding variables as part of the data analysis.
3. Tempering the causal strength of claims in light of the inherent challenges in (1) and (2).

> Furthermore, note that (1) implicates a careful data collection planning.

iClicker Question

Answer TRUE or FALSE:

Identifying potential confounding variables is not an integral part of a careful data collection planning in an observational study.

A. TRUE

B. FALSE

Answer

- It is **FALSE**.
- Since we cannot randomize our treatments of interest in an observational study, we cannot automatically eliminate the effect of all potential confounders.
- Therefore, it is essential to carefully design our observational study with the help of people with proper expertise. This help would allow us to identify potential confounding variables.

1.2. Where are observational studies in a Data Science pipeline?

Observational studies are heavily used in pharmaco-epidemiology where we can find massive healthcare databases for population health research. For instance, insurance companies and healthcare systems collect this class of data in the US. On the other hand, countries like Canada use their national healthcare system to gather this data.

In addition to statistical methodology issues, this sort of work heavily involves:

- Database and data wrangling issues (defining and extracting populations/samples/variables).
- Privacy, ethics, and security (you already addressed these matters in DSCI 541).

Therefore, we generally have an outcome Y (which could be continuous OR discrete) subject to a variable of interest X and many confounders C_j which will vary in their nature: continuous OR discrete.

For example

For instance, in a pharmaco-epidemiology problem:

- Y is a binary indicator of bad disease state two years after initial diagnosis.
- X is a binary indicator of initiated drug A versus drug B within three months of initial diagnosis.
- C_j is all sorts of demographic variables and health-status variables.

> We aspire to use these observational data to determine what we would see if we could do a randomized trial with this X and Y.

Heads-up

Merely addressing whether X is associated with Y IS NOT HELPFUL ANYMORE. In the previous pharmaco-epidemiology problem, the relevant policy question is:

> Would fewer patients reach the bad disease state in a world where all were given drug B compared to a world where all were given drug A?

2. The Western Collaborative Group Study Data

- As in the case of A/B testing data, pharmaco-epidemiology data is hard to come by!
- Hence, we will work with a more traditional biostatistics example from **Vitinghoff et al. (2012)**.
- This dataset is called the Western Collaborative Group Study (WCGS) data.

Summary of Dataset

The WCGS data is a prospective cohort study that recruited middle-aged men between the ages of 39 and 59 who were employees of ten Californian companies. The data was collected on 3154 subjects between 1960 and 1961.

> Heads-up: A prospective cohort study involves following n subjects over time. Certain confounding variables can group these subjects. Nonetheless, these groups will vary by a certain variable of interest (i.e., the X as a factor-type variable with certain levels of interest).

2.1. Data Wrangling

- The data `wcgs.dta` is in Stata format.
- We will use the function `read.dta()` from `foreign` as a tibble.
- We will eliminate those rows with missing data via `na.omit()` (missing data will not be the scope of today's lecture!).
- We end up with $n = 3,101$ subjects.

Read Data in R!

R Code Start Over Run Code

```r
wcgs <- as_tibble(read.dta("data/wcgs.dta"))
wcgs <- na.omit(wcgs)
wcgs
```

```
# A tibble: 3,101 × 22
     age arcus behpat   bmi chd69  chol   dbp dibpat height    id lnsbp lnwght

 1    50     1 A1      31.3 No      249    90 Type A     67  2343  4.88   5.30
 2    51     0 A1      25.3 No      194    74 Type A     73  3656  4.79   5.26
 3    59     1 A1      28.7 No      258    94 Type A     70  3526  5.06   5.30
 4    51     1 A1      22.1 No      173    80 Type A     69 22057  4.84   5.01
 5    44     0 A1      22.3 No      214    80 Type A     71 12927  4.84   5.08
 6    47     0 A1      27.1 No      206    76 Type A     64 16029  4.75   5.06
 7    40     0 A1      23.2 No      190    78 Type A     70  3894  4.80   5.09
 8    41     0 A1      23.0 No      212    84 Type A     70 11389  4.87   5.08
 9    50     1 A1      27.2 No      130    70 Type A     71 12681  4.72   5.27
10    43     0 A1      28.4 No      233    80 Type A     68 10005  4.79   5.23
# ℹ 3,091 more rows
# ℹ 10 more variables: ncigs , sbp , smoke , t1 ,
#   time169 , typchd69 , uni , weight , wghtcat ,
#   agec
```

Variables in `wcgs`

Our raw data `wcgs` has 22 demographic and health-related variables in total. However, our statistical modelling will focus on the following ones:

- `dibpat`: Dichotomous behaviour pattern, a factor-type variable with two levels (`Type A` and `Type B`).
- `age`: Subject's age in years (a count-type variable).
- `bmi`: Subject's body mass index (BMI) in kg/m^2 (a continuous-type variable).

Variables in `wcgs` (Continued)

- `chol`: Subject's cholesterol levels in $\mathrm{mg}/100\ \mathrm{ml}$ (a continuous-type variable).
- `smoke`: Whether the subject smokes or not, a factor-type variable with two levels (`Yes` and `No`).
- `chd69`: Whether the subject experienced a coronary heart disease (CHD) event, a factor-type variable with two levels (`Yes` and `No`).

Make our `training_data`

These variables are stored in `training_data`.

R Code Start Over Run Code

```
training_data <- wcgs %>% select(dibpat, age, bmi, chol, smoke, chd69)
training_data
```

```
# A tibble: 3,101 × 6
   dibpat   age   bmi  chol smoke chd69

 1 Type A    50  31.3   249 Yes   No
 2 Type A    51  25.3   194 Yes   No
 3 Type A    59  28.7   258 No    No
 4 Type A    51  22.1   173 No    No
 5 Type A    44  22.3   214 No    No
 6 Type A    47  27.1   206 Yes   No
 7 Type A    40  23.2   190 No    No
 8 Type A    41  23.0   212 Yes   No
 9 Type A    50  27.2   130 No    No
10 Type A    43  28.4   233 Yes   No
# ℹ 3,091 more rows
```

2.2. Main Statistical Inquiry

We are interested in the following: does a `Type A` behaviour pattern (`dibpat`) LEAD to a CHD event (`chd69`)? From **Brand et al. (1976)**, the levels of `dibpat` are defined as follows:

> *Methods for the assessment of the dichotomous behavior pattern by a structured psychological interview in the WCGS have been described elsewhere. Type A behavior is characterized by enhanced aggressiveness and competitive drive, preoccupation with deadlines, and chronic impatience and sense of time urgency, in contrast with the more relaxed and less hurried Type B behavior pattern.*

Hence

We have an X variable, a Y variable, and different potential C_j variables. Which variables are we focusing on?

- Y is binary indicator of coronary heart disease (`chd69`).
- X is binary indicator of `Type A` behaviour (`dibpat`).

2.3. Exploratory Data Analysis

Let us do a quick exploratory data analysis (EDA).

R Code Start Over Run Code

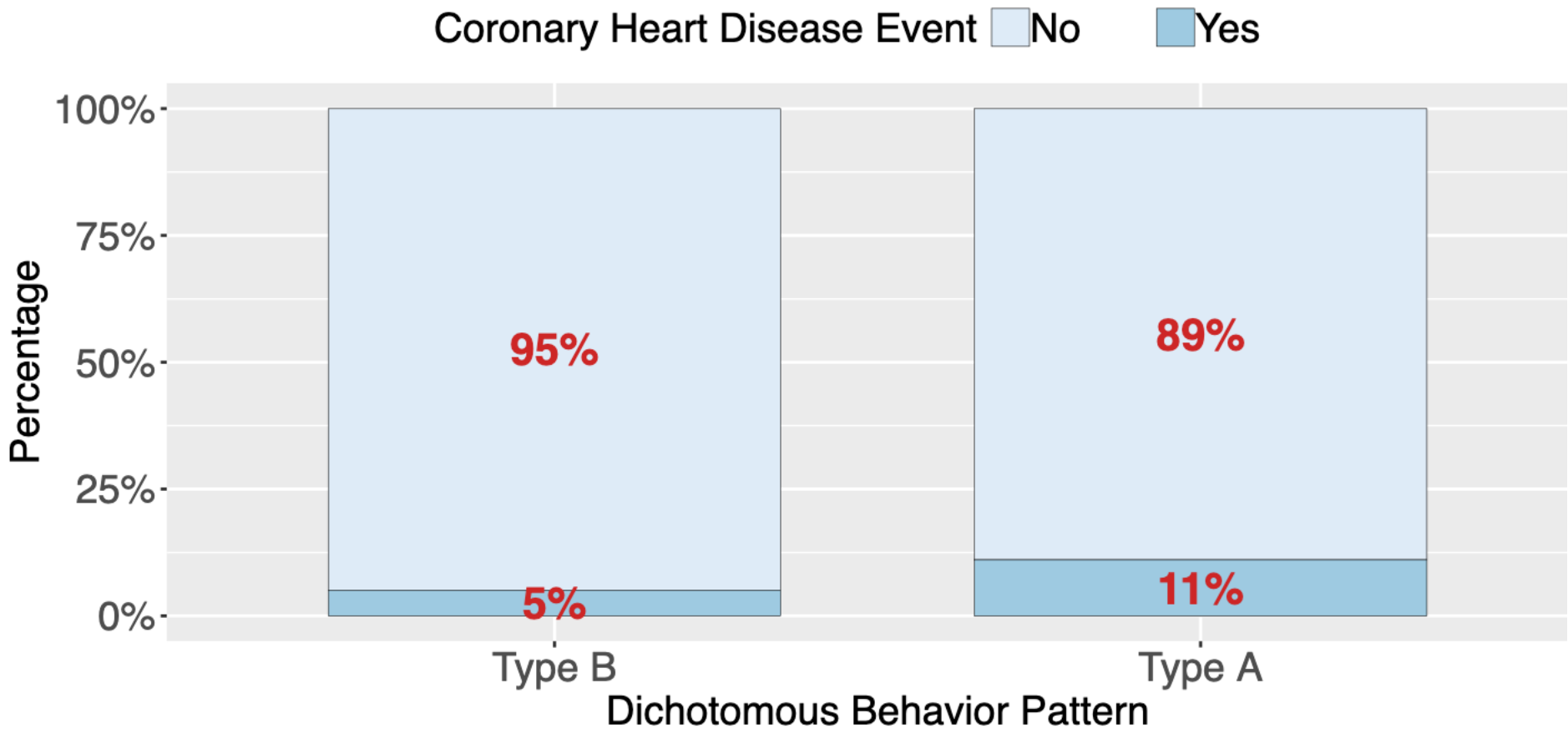
```
wcgs_prop_summary <- as.data.frame(xtabs(
  ~ dibpat + chd69,
  training_data
) / rowSums(xtabs(
  ~ dibpat + chd69,
  training_data
)), responseName = "prop")

wcgs_data_stacked_bars <- ggplot(wcgs_prop_summary, aes(x = dibpat, y = prop,
  geom_bar(stat = "identity", width = 0.7, colour = "black", linewidth = 0.1)
  geom_text(aes(label = ifelse(prop >= 0.05, paste0(sprintf("%.0f", prop*100)
            position = position_stack(vjust = 0.5), colour = "firebrick3", fo
  scale_y_continuous(labels = percent_format()) +
  labs(y = "Percentage", x = "Dichotomous Behavior Pattern", fill = "") +
  ggtitle("")  +
  theme(axis.text.x = element_text(size = 18, angle = 0),
        axis.text.y = element_text(size = 18, angle = 0),
        axis.title = element_text(size = 18),
        legend.text = element_text(size = 18, margin = margin(r = 1, unit = "
        legend.title = element_text(size = 18),
        legend.position = "top") +
  guides(fill = guide_legend(title = "Coronary Heart Disease Event")) + scale
```

Plot in R!

R Code Start Over Run Code

```
wcgs_data_stacked_bars
```


Coronary Heart Disease Event
No
Yes
100%
75%
50%
25%
0%
Percentage
95%
5%
89%
11%
Type B
Type A
Dichotomous Behavior Pattern

3. Key Statistical Concepts

For the ith subject, the response `chd69` is defined as:

$$Y_i = \begin{cases} 1 & \text{if the } i\text{th subject experienced a CHD event,} \\ 0 & \text{otherwise.} \end{cases}$$

The variable `dibpat` is defined as:

$$X_i = \begin{cases} 1 & \text{if the } i\text{th subject has a Type A behaviour pattern,} \\ 0 & \text{otherwise.} \end{cases}$$

3.1. Log-odds

Recall the "1" category in Y_i is referred as *success*.
Note each Y_i is a Bernoulli trial whose probability of success is π_i, i.e.,
$Y_i \sim \text{Bernoulli}(\pi_i)$
We are already familiar with the log-odds:
$$\log\left(\frac{\pi_i}{1-\pi_i}\right)$$
The log-odds is to binary variables what correlation is to continuous variables.

3.2. Estimator of the Log-Odds for One Group

In a simple case with a sample size n, if we just have one variable (e.g., CHD or not), we can estimate the log-odds by just taking the logarithm of the ratio of the number of patients with CHD to those without CHD:

$$\text{log-odds} = \log\left(\frac{n_{\text{yes}}/n}{n_{\text{no}}/n}\right) = \log\left(\frac{n_{\text{yes}}}{n_{\text{no}}}\right)$$

3.3. Estimator of the Log-Odds Ratio for Two Groups

Now, let us add the variable X to our statistical metrics besides the outcome Y.
Suppose both X and Y are binary variables.
Then, the absolute frequencies by each combination of factor levels can be put in a contingency table as follows:

	$Y = 1$	$Y = 0$
$X = 1$	$n_{X=1,Y=1}$	$n_{X=1,Y=0}$
$X = 0$	$n_{X=0,Y=1}$	$n_{X=0,Y=0}$

> Heads-up: A contingency table (i.e., cross-tabulation) is merely a way to summarize the absolute frequencies between two categorical variables.

Definition of odds ratio (OR)

The odds ratio (OR) measures the association/causation between the binary exposure variable X and the binary outcome Y.
It will indicate the odds that the outcome Y will happen given that the exposure X is present compared to the odds the outcome Y will happen if the exposure X is absent.

Formula for odds ratio (OR)

It can be represented as follows:

$$\text{OR} = \frac{\frac{n_{X=1,Y=1}/n}{n_{X=1,Y=0}/n}}{\frac{n_{X=0,Y=1}/n}{n_{X=0,Y=0}/n}} = \frac{\frac{n_{X=1,Y=1}}{n_{X=1,Y=0}}}{\frac{n_{X=0,Y=1}}{n_{X=0,Y=0}}} = \frac{n_{X=1,Y=1} \times n_{X=0,Y=0}}{n_{X=1,Y=0} \times n_{X=0,Y=1}}$$

Interpretation

The OR can be interpreted as follows:

- $\mathrm{OR} = 1$ indicates the binary exposure X DOES NOT AFFECT the odds of the binary outcome Y.
- $\mathrm{OR} > 1$ indicates the binary exposure X IS ASSOCIATED WITH (**OR CAUSES!**) higher odds of the binary outcome Y.
- $\mathrm{OR} < 1$ indicates the binary exposure X IS ASSOCIATED WITH (**OR CAUSES!**) lower odds of the binary outcome Y.

Estimator of the log-OR

Similar to the estimator of the log-odds for one group, we can also have an estimator of the log-OR as follows:

$$\hat{\text{log-OR}} = \log\left(\frac{\frac{n_{X=1,Y=1}}{n_{X=1,Y=0}}}{\frac{n_{X=0,Y=1}}{n_{X=0,Y=0}}}\right) = \log\left(\frac{n_{X=1,Y=1}}{n_{X=1,Y=0}}\right) - \log\left(\frac{n_{X=0,Y=1}}{n_{X=0,Y=0}}\right)$$

$$\begin{aligned}\hat{\text{log-OR}} = &\log(n_{X=1,Y=1}) - \log(n_{X=1,Y=0}) \\ &- \left[\log(n_{X=0,Y=1}) - \log(n_{X=0,Y=0})\right]\end{aligned}$$

Approximate Distribution of Odds-Ratio

This estimator is approximately Normal with a large enough sample size n. Its standard error (SE) is:

$$\text{SE} = \sqrt{\frac{1}{n_{X=1,Y=1}} + \frac{1}{n_{X=1,Y=0}} + \frac{1}{n_{X=0,Y=1}} + \frac{1}{n_{X=0,Y=0}}}$$

> We can set up the following R functions with a contingency `table` as an input following the previous formulas.

The last function, `logOR_z`, is the test statistic `logOR_z` associated to the hypothesis:

H_0: log-OR $= 0$

H_a: log-OR $\neq 0$

Failing to reject H_0 gives statistical evidence that the binary exposure X DOES NOT AFFECT the odds of the binary outcome Y with a significance level α.

Functions in R!

R Code Start Over Run Code

```r
logOR_est <- function(table) {
  log(table[2, 2]) - log(table[2, 1]) - (log(table[1, 2]) - log(table[1, 1]))
}

logOR_se <- function(table) {
  sqrt(sum(1 / table))
}

logOR_z <- function(table) {
  logOR_est(table) / logOR_se(table)
}
```

3.4. Applying the log-OR to the WCGS Case

Let us apply the concept of the log-OR to our training data for our two target variables: the exposure X called behaviour pattern (`dibpat`) and the outcome Y called CHD event (`chd69`).
Function `table()` computes the corresponding contingency tables.

R Code Start Over Run Code

```r
cont_table <- table(training_data$dibpat, training_data$chd69)
cont_table
```

```
          No  Yes
Type B 1459   78
Type A 1391  173
```

Computing log-odds ratio

Then we use the function `logOR_est()`.

R Code Start Over Run Code

```
round(logOR_est(cont_table), 3)
```

```
[1] 0.844
```

We can obtain the interpretation of the OR by exponentiating the log-OR.

R Code Start Over Run Code

```
round(exp(logOR_est(cont_table)), 3)
```

```
[1] 2.326
```

How to interpret?

An $OR > 1$ indicates the Type A behaviour pattern IS ASSOCIATED WITH (OR POSSIBLY CAUSES!) higher odds of having a CHD event.

Standard Error and Test Statistic

We will also obtain the corresponding SE and test statistic associated with the estimated log-OR.

R Code Start Over Run Code

```
round(logOR_se(cont_table), 3)
```

```
[1] 0.141
```

R Code Start Over Run Code

```
round(logOR_z(cont_table), 3)
```

```
[1] 5.969
```

Relationship to Logistic Regression Model

Finally, it can be proved that the previous estimated log-OR, SE, and test statistic can be obtained with a Binomial Logistic regression model with a single binary regressor X.

We fit this model and call it `initial_bin_log`.

R Code Start Over Run Code

```
initial_bin_log <- glm(chd69 ~ dibpat, data = training_data, family = binomial
tidy(initial_bin_log, conf.int = 0.95) %>% mutate_if(is.numeric, round, 3)
```

```
# A tibble: 2 × 7
  term         estimate std.error statistic p.value conf.low conf.high

1 (Intercept)    -2.93      0.116    -25.2        0   -3.16      -2.71
2 dibpatType A    0.844     0.141      5.97       0    0.571      1.13
```

Any Confounders?

We called the previous model `initial_bin_log`, assuming we just look at X (`dibpat`) and Y (`chd69`).
The adjective "*initial*" indicates this model is not accounting for all possible confounders C_j, thus we can only infer association.
However, this DOES NOT ADDRESS the main statistical inquiry of causality between X and Y.
Hence, we need to look at what confounders we might include in our analysis.

4. Stratified Analysis for Causality

Stratification implicates slicing our data based on different confounding variables. This will yield a stratified analysis for causality.

> Heads-up: Often, one sees the term stratification as a short-hand for stratified sampling (i.e., collecting subsamples within each stratum). However, this will not be the case in this analysis.

4.1. Stratum-Specific Inference with a Single Counfounder

In a stratified analysis for causality, we can make bins with a given continuous confounder.
Then, we will stratify our sampled subjects by each confounder bin according to their observed value.
In an observational study, this is the PROXY to blocking in an experiment.
A fair stratifying practice with a continuous confounder is to build quartile stratum, i.e., split the ordered data into four parts (as we do in boxplots!).

age Variable as Counfounder

We will do it for age in training_data.

R Code Start Over Run Code

```
training_data <- training_data %>%
  mutate(age_bins = cut(age, breaks = c(min(age), quantile(age, (1:3) / 4), ma
levels(training_data$age_bins)
```

```
[1] "[39,42]" "(42,45]" "(45,50]" "(50,59]"
```

R Code Start Over Run Code

```r
head(training_data)
```

```
# A tibble: 6 × 7
  dibpat   age   bmi  chol smoke chd69 age_bins

1 Type A    50  31.3   249 Yes   No    (45,50]
2 Type A    51  25.3   194 Yes   No    (50,59]
3 Type A    59  28.7   258 No    No    (50,59]
4 Type A    51  22.1   173 No    No    (50,59]
5 Type A    44  22.3   214 No    No    (42,45]
6 Type A    47  27.1   206 Yes   No    (45,50]
```

Then

As we did it previously with the overall contingency table, we will build four specific contingency tables by each `age` stratum of the X `dibpat` versus the Y `chd69`.

R Code Start Over Run Code

```r
attach(training_data)
table(dibpat, chd69, age_bins)
```

```
, , age_bins = [39,42]

        chd69
dibpat    No Yes
  Type B 504  17
  Type A 432  30

, , age_bins = (42,45]

        chd69
dibpat    No Yes
  Type B 326  15
  Type A 262  21

, , age_bins = (45,50]

        chd69
dibpat    No Yes
  Type B 340  21
  Type A 332  49

, , age_bins = (50,59]

        chd69
dibpat    No Yes
  Type B 289  25
  Type A 365  73
```

Estimated log-OR by age Stratum

We obtain the estimate log-OR by age stratum.

R Code Start Over Run Code

```
log_OR_age <- matrix(round(apply(table(dibpat, chd69, age_bins), 3, logOR_est)
colnames(log_OR_age) <- levels(training_data$age_bins)
log_OR_age
```

```
     [39,42] (42,45] (45,50] (50,59]
[1,]   0.722   0.555   0.871   0.838
```

Estimated OR by age Stratum

If we exponentiate the previous estimates, we obtain the corresponding ORs by age stratum. Note that all ORs are larger than 1.

R Code Start Over Run Code

```
OR_age <- matrix(round(exp(apply(table(dibpat, chd69, age_bins), 3, logOR_est)
colnames(OR_age) <- levels(training_data$age_bins)
OR_age
```

```
     [39,42] (42,45] (45,50] (50,59]
[1,]   2.059   1.742    2.39   2.312
```

Estimated SE by age Stratum

We also obtain the SEs and test statistics by age stratum.

R Code Start Over Run Code

```
sd_log_OR_age <- matrix(round(apply(table(dibpat, chd69, age_bins), 3, logOR_s
colnames(sd_log_OR_age) <- levels(training_data$age_bins)
sd_log_OR_age
```

```
     [39,42] (42,45] (45,50] (50,59]
[1,]   0.311   0.348   0.272   0.245
```

Rejection Criteria

With $\alpha = 0.05$ and a two-sided test
H_0: $\text{log-OR} = 0$
H_a: $\text{log-OR} \neq 0$,
our quantile cuttoff value $z_{1-\alpha/2} = 1.96$.

Therefore

We have statistical evidence to reject H_0 (recall H_0 indicates the binary exposure X DOES NOT AFFECT the odds of the binary outcome Y) for all age stratum except for `(42,45]`.

Z-Statistic by age Stratum

R Code Start Over Run Code

```
test_log_OR_age <- matrix(round(apply(table(dibpat, chd69, age_bins), 3, logOR
colnames(test_log_OR_age) <- levels(training_data$age_bins)
test_log_OR_age
```

```
     [39,42] (42,45] (45,50] (50,59]
[1,]   2.325   1.594   3.203   3.424
```

Summary

What did we just do in this whole analysis?

Assuming age is the only possible confounder in this population, we take away its potential of biasing our effect estimates of X on Y.

Ideally, this stratification technique will yield more accurate estimates in the form of log-ORs.

> Heads-up: You can make different age strata if necessary. This will depend on what is specifically of interest for your main statistical inquiry.

4.2. Stratum-Specific Causal Inference with Multiple Counfounders

Now that we have done this stratified analysis by `age`, it is also possible to expand it to multiple confounders C_j.
Hence, let us incorporate `bmi` as another confounder in the form of four strata by quartile.

R Code Start Over Run Code

```
training_data <- training_data %>%
  mutate(bmi_bins = cut(bmi, breaks = c(min(bmi), quantile(bmi, (1:3) / 4), ma
levels(training_data$bmi_bins)
```

```
[1] "[11.2,23]"   "(23,24.4]"   "(24.4,25.8]" "(25.8,38.9]"
```

All Possible Combinations of Stratum Level

If we stratify by age group (4 levels), smoking status (2 levels), BMI quartile (4 levels); we will end with following possible combinations of stratum levels: $4 \times 2 \times 4 = 32$

Code in R!

R Code Start Over Run Code

```r
head(training_data)
```

```
# A tibble: 6 × 8
  dibpat   age   bmi  chol smoke chd69 age_bins bmi_bins

1 Type A    50  31.3   249 Yes   No    (45,50]  (25.8,38.9]
2 Type A    51  25.3   194 Yes   No    (50,59]  (24.4,25.8]
3 Type A    59  28.7   258 No    No    (50,59]  (25.8,38.9]
4 Type A    51  22.1   173 No    No    (50,59]  [11.2,23]
5 Type A    44  22.3   214 No    No    (42,45]  [11.2,23]
6 Type A    47  27.1   206 Yes   No    (45,50]  (25.8,38.9]
```

32 specific contingency tables

Then, we build 32 specific contingency tables by each stratum of the X `dibpat` versus the Y `chd69`.

R Code Start Over Run Code

```
attach(training_data)
mult_cont_tables <- table(dibpat, chd69, age_bins, smoke, bmi_bins)
```

```
The following objects are masked from training_data (pos = 3):

    age, age_bins, bmi, chd69, chol, dibpat, smoke
```

The object `mult_cont_tables` is multidimensional as follows:

R Code Start Over Run Code

```
dim(mult_cont_tables)
```

```
[1] 2 2 4 2 4
```

One of Possible Contingency Tables

We can access a given contingency table with the syntax below (for instance, for stratum `(45,50]` in `age`, `Yes` in `smoke`, and `(25.8,38.9]` in `bmi`).

R Code Start Over Run Code

```
mult_cont_tables[ , , 3, 2, 4]
```

```
        chd69
dibpat   No Yes
  Type B 30   4
  Type A 36   5
```

> Note, that is also possible to check how many subjects we have in each one stratum.

Sample Size in Each Strata

R Code Start Over Run Code

```r
# Subjects in each one of the strata
n_strata <- apply(mult_cont_tables, 3:5, sum)
n_strata
```

```
, , bmi_bins = [11.2,23]

         smoke
age_bins   No Yes
  [39,42] 119 147
  (42,45]  79  84
  (45,50]  69 121
  (50,59]  76 115

, , bmi_bins = (23,24.4]

         smoke
age_bins   No Yes
  [39,42] 136 107
  (42,45]  64  73
  (45,50]  96  81
  (50,59]  93  99

, , bmi_bins = (24.4,25.8]

         smoke
age_bins   No Yes
  [39,42] 136 115
```

```
  (42,45]  97  72
  (45,50] 107  83
  (50,59]  99  69

, , bmi_bins = (25.8,38.9]

         smoke
age_bins   No Yes
  [39,42] 131  92
  (42,45]  93  62
  (45,50] 110  75
  (50,59] 124  77
```

For each one of these 32 strata

We can obtain their corresponding estimates for the log-ORs, their SEs, and test statistics.
Note that some strata show `NaN`, `Inf`, or `-Inf`; this happens because some contingency tables had cells with the value of zero as an absolute frequency. Hence, computations are not possible.

> Heads-up: We can already see we are running into issues. The more variables we stratify, the smaller the subsample size within each stratum. Therefore, we might not have enough sampled subjects by strata!

Log-odds Ratio

R Code Start Over Run Code

```r
est_strata_or <- apply(mult_cont_tables, 3:5, logOR_est)
round(est_strata_or, 3)
```

```
, , bmi_bins = [11.2,23]

         smoke
age_bins     No    Yes
  [39,42]   Inf -0.294
  (42,45]   NaN  0.583
  (45,50] 1.634  0.182
  (50,59] 0.611  0.400

, , bmi_bins = (23,24.4]

         smoke
age_bins     No   Yes
  [39,42] 1.099 0.775
  (42,45]   NaN 0.405
  (45,50] 0.321 2.253
  (50,59] 0.635 0.361

, , bmi_bins = (24.4,25.8]

         smoke
age_bins      No   Yes
  [39,42]  0.972 1.643
  (42,45]   -Inf 0.405
```

```
  (45,50] -0.405 0.889
  (50,59]  0.981 0.734

, , bmi_bins = (25.8,38.9]

         smoke
age_bins     No   Yes
  [39,42] 0.209 0.740
  (42,45] 1.245 0.511
  (45,50] 2.099 0.041
  (50,59] 1.887 0.934
```

Standard Error

R Code Start Over Run Code

```r
se_strata <- apply(mult_cont_tables, 3:5, logOR_se)
round(se_strata, 3)
```

```
, , bmi_bins = [11.2,23]

         smoke
age_bins     No   Yes
  [39,42]   Inf 0.750
  (42,45]   Inf 0.940
  (45,50] 1.183 0.786
  (50,59] 1.247 0.577

, , bmi_bins = (23,24.4]

         smoke
age_bins     No   Yes
  [39,42] 1.168 0.758
  (42,45]   Inf 1.248
  (45,50] 0.739 1.072
  (50,59] 0.865 0.554

, , bmi_bins = (24.4,25.8]

         smoke
age_bins     No   Yes
  [39,42] 0.884 1.097
  (42,45]   Inf 0.694
```

```
  (45,50] 0.934 0.627
  (50,59] 0.710 0.857

, , bmi_bins = (25.8,38.9]

         smoke
age_bins     No   Yes
  [39,42] 0.836 0.892
  (42,45] 0.865 0.779
  (45,50] 1.079 0.715
  (50,59] 0.771 0.632
```

Which strata have no empty cells?

R Code Start Over Run Code

```r
no_empty_strata <- (as.vector(se_strata) < Inf)
sum(no_empty_strata)
```

```
[1] 28
```

Find 95% Confidence Intervals (CIs)

For those 28 strata where computations were possible, we will obtain their corresponding 95% confidence intervals (CIs).
Plotting these CIs will identify how many log-OR estimates were statistically significant out of those 28.

Code in R!

R Code Start Over Run Code

```
CI_strata <- matrix(0.0, length(se_strata), 2)
sig_strata <- rep(FALSE, length(se_strata))

z_alpha <- qnorm(0.975)

# Compute 95% CIs
for (i in 1:length(no_empty_strata)) {
  if (no_empty_strata[i]) {
    CI_strata[i, ] <- est_strata_or[i] + z_alpha * c(-1, 1) * se_strata[i]
    sig_strata[i] <- ((CI_strata[i, 1] > 0) || (CI_strata[i, 2] < 0))
  }
}
```

Confidence Intervals (95%)

R Code Start Over Run Code

```
head(CI_strata)
```

```
           [,1]     [,2]
[1,]  0.0000000 0.000000
[2,]  0.0000000 0.000000
[3,] -0.6844251 3.952686
[4,] -1.8330703 3.054888
[5,] -1.7640614 1.175752
[6,] -1.2600192 2.426312
```

Code in R!

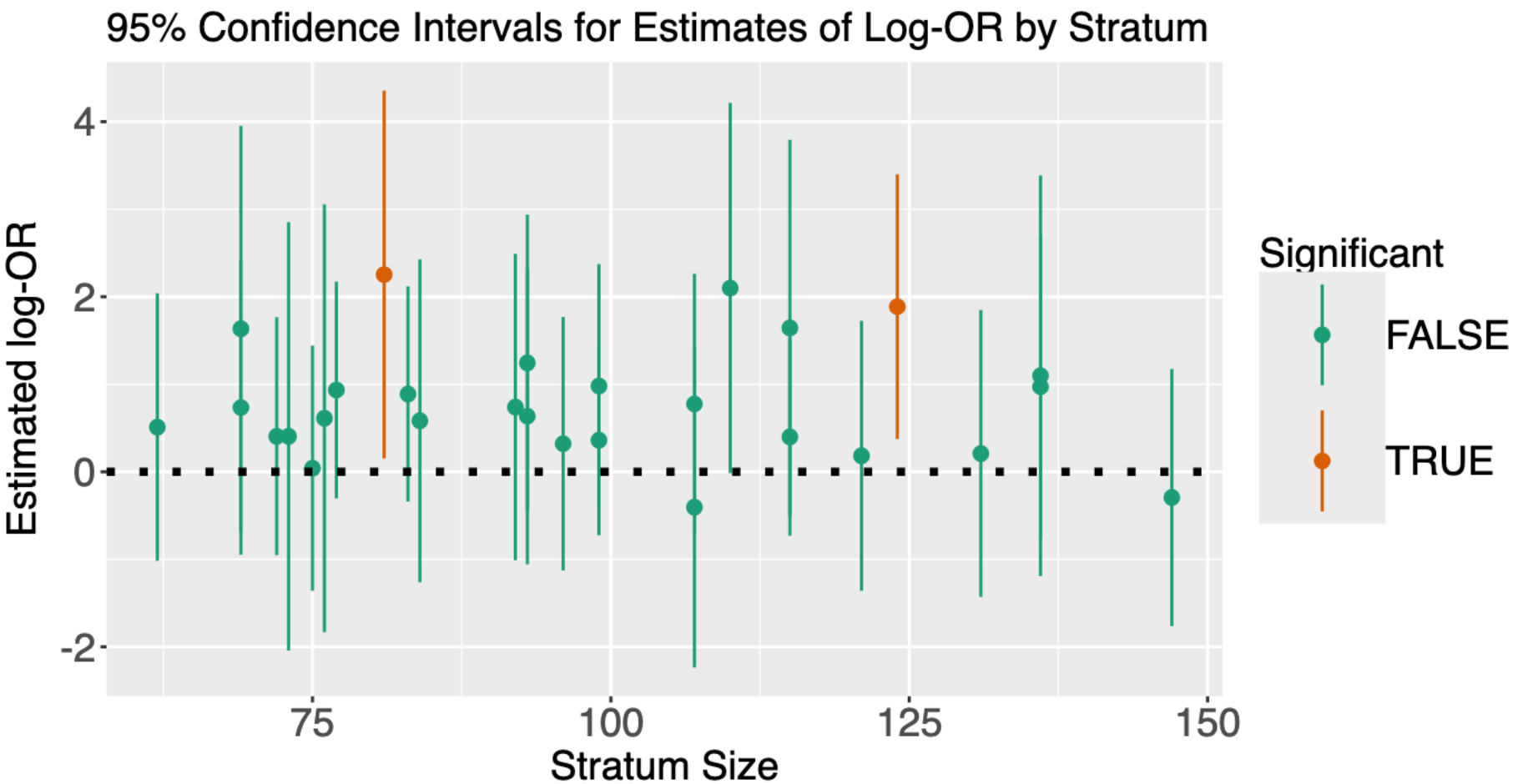
R Code Start Over Run Code

```r
strata_log_OR_95_CI <- data.frame(
  logORest = est_strata_or[no_empty_strata],
  no_empty_strata = n_strata[no_empty_strata],
  CI = CI_strata[no_empty_strata, ], Significant = sig_strata[no_empty_strata
) %>%
  ggplot(aes(x = no_empty_strata, y = logORest, colour = Significant)) +
  geom_pointrange(aes(ymin = CI.1, ymax = CI.2)) +
  ylab("Estimated log-OR") +
  xlab("Stratum Size") +
  geom_hline(yintercept = 0, linetype = "dotted", linewidth = 1.3) +
  theme(
    plot.title = element_text(size = 18),
    axis.text = element_text(size = 18),
    axis.title = element_text(size = 18),
    legend.position = "right",
    legend.title = element_text(size = 18, margin = margin(r = 1, unit = "cm"
    legend.text = element_text(size = 18, margin = margin(r = 1, unit = "cm")
    legend.key.size = unit(2, 'cm')
  ) +
  ggtitle("95% Confidence Intervals for Estimates of Log-OR by Stratum") +
  scale_color_brewer(palette = "Dark2")
```

Plotting CIs!

R Code Start Over Run Code

```
strata_log_OR_95_CI
```


95% Confidence Intervals for Estimates of Log-OR by Stratum
Estimated log-OR
4
2
0
-2
75
100
125
150
Stratum Size
Significant
FALSE
TRUE

What is the main takeaway from this CI plot?

- Only two strata had significant results in terms of the estimated log-OR!
- Moreover, we can see some other strata were close to being significant (but not yet!).

In-Class Question

Is there anything invalid from a statistical standpoint? Or, if it is not invalid, is there anything that we might want to do differently?

Answer

When we start slicing our data, in this case, we have 32 different different strata. These are the facts and struggles:

- By making stratum-specific inference with multiple confounders, we aim to infer causality between X and Y.
- However, many of these strata do not have enough observations to estimate a log-OR!
- Hence, running a separate analysis within each stratum hinders your power.

Answer (Continued)

- There is another big drawback: we are doing multiple testing. Hence, these raw inferential results would need the corresponding adjustment (which would hinder power even more!).
- If we want to do things differently, we need a more clever model: overall Binomial Logistic regression.

4.3. Causal Inference via an Overall Logistic Regression

The overall logistic regression would be our more clever approach. It will include the binary variables X and Y and the corresponding confounders C_j AS STRATA. This is our proxy for blocking in a randomized experiment!

> Note all regressors are modelled as standalone variables.

R Code Start Over Run Code

```r
bin_log_model <-  glm(chd69 ~ dibpat + age_bins + bmi_bins + smoke, family = "
tidy(bin_log_model, conf.int = 0.95) %>% mutate(exp.estimate = exp(estimate))
  mutate_if(is.numeric, round, 3)
```

```
# A tibble: 9 × 8
  term      estimate std.error statistic p.value conf.low conf.high exp.estimate

1 (Interce…   -4.24      0.246   -17.2     0       -4.74     -3.77         0.014
2 dibpatTy…    0.73      0.144     5.08    0        0.452     1.01         2.08
3 age_bins…    0.19      0.23      0.826   0.409   -0.266     0.638        1.21
4 age_bins…    0.687     0.197     3.48    0        0.304     1.08         1.99
5 age_bins…    1.02      0.187     5.47    0        0.663     1.40         2.78
6 bmi_bins…    0.413     0.208     1.98    0.048    0.006     0.826        1.51
7 bmi_bins…    0.569     0.205     2.78    0.005    0.171     0.975        1.77
8 bmi_bins…    0.796     0.199     4.01    0        0.412     1.19         2.22
9 smokeYes     0.704     0.139     5.05    0        0.433     0.979        2.02
```

- Using the same log-OR plot with the 95% CIs, we illustrate the 95% CI for the regression coefficient `estimate dibpatType A` (as dashed horizontal blue lines).
- The `estimate` corresponds to the solid horizontal blue line (it is significant!).
- If we compare this blue 95% CI with the other 26 95% CIs, we get a more precise estimate of X with this overall Binomial Logistic regression model.

Code in R!

R Code Start Over Run Code

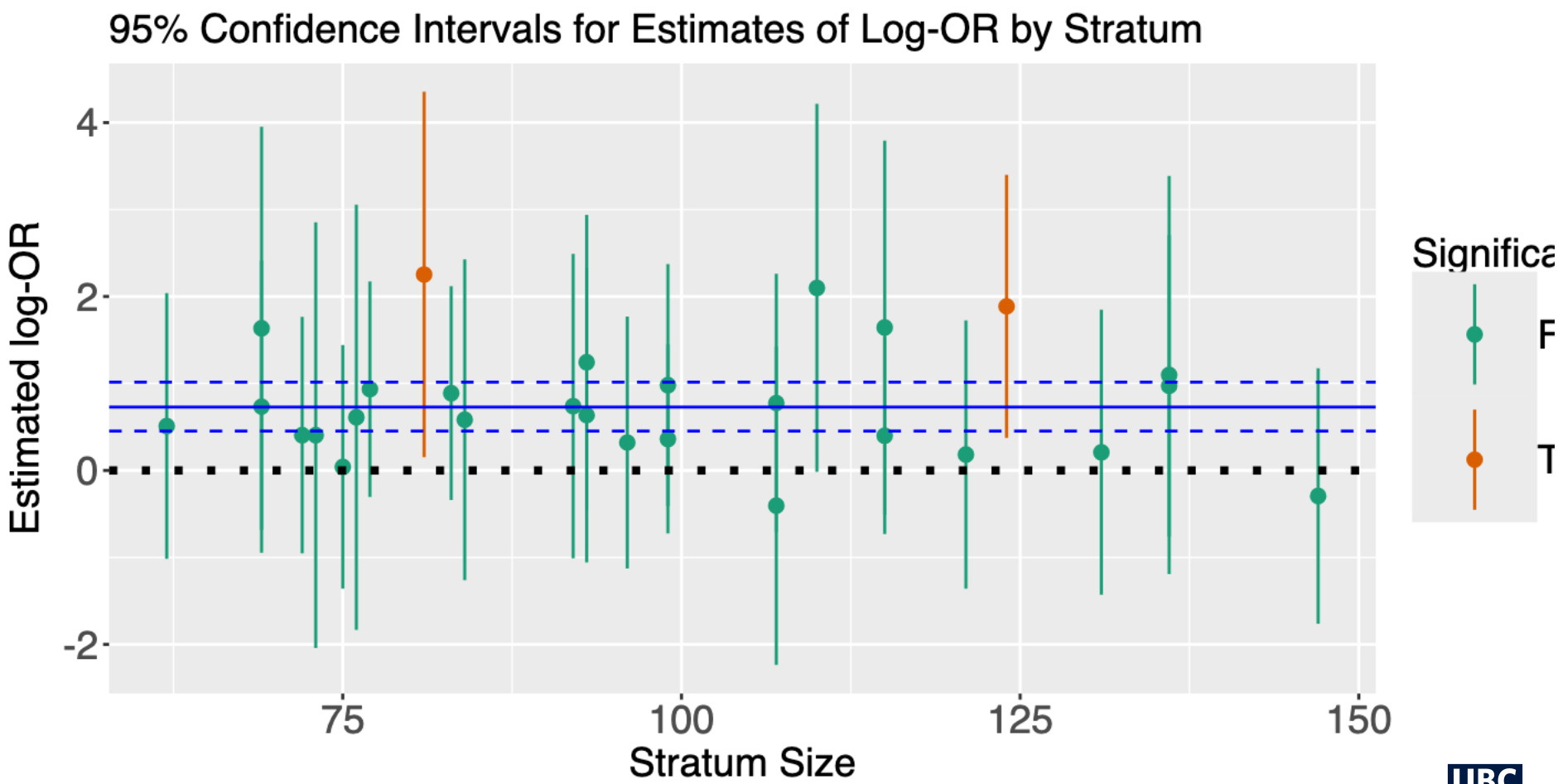
```
options(warn = -1)
plot_ci <- strata_log_OR_95_CI + geom_hline(yintercept = bin_log_model$coeff[2
  geom_hline(yintercept = confint(bin_log_model)[2, ], linetype = "dashed", co
  scale_color_brewer(palette = "Dark2")
```

```
Waiting for profiling to be done...


Scale for colour is already present.
Adding another scale for colour, which will replace the existing scale.
```

R Code Start Over Run Code

```
plot_ci
```



Conclusion so far

If we *REALLY* want to put a causal spin on our estimate for the X coefficient, what must we assume?

By controlling for possible confounders such as age, BMI, and smoking status in our regression model; we are assuming that no matter what age, smoking status, or BMI this population of interest has, all subjects have the same effect of X over Y:

> *Being a middle-aged man between the ages of 39 and 59 (and an employee of a Californian company) with a Type A behaviour (when compared to Type B) LEADS to being 2.075 times more likely to experience a CHD event.*

However

unlike a stratum-specific inference with multiple confounders, this class of causal conclusions must take various assumptions into account. Hence, bearing in mind that we fit:

```
glm(chd69 ~ dibpat + age_bins + bmi_bins + smoke, family = "binomial")
```

We will explore these assumptions.

4.4. Assumptions for the Causal Overall Binomial Logistic Regression

The big idea of our `bin_log_model` is that this model is enough to assess causality of X on Y. Nevertheless, this idea involves the following strong assumptions:

1. There is a simple/smooth structure in how the Y-specific log-OR varies across the strata.
2. The strength of the (X, Y) association within each of the 32 strata defined by the confounders is the same (i.e., we DO NOT HAVE any double interactions between X and each confounder).
3. This example assumes we have included all of our confounding variables in our regression model. Hence, there are only three for this model: age, smoking status, and BMI.

4.4.1. Stress test of the FIRST assumption (simple structure)

We will use model selection to assess this assumption. Since our `bin_log_model` is a maximum likelihood-based approach, we can then run a likelihood ratio test (LRT) to assess goodness of fit between two models: model 1 and model 2:

- model 1: It is a simpler model. This model is nested in model 2.
- model 2: It is a more complex model.

What are are the hypotheses?

The hypotheses are:

$$H_0: \text{model 1 fits the data better than model 2}$$
$$H_a: \text{otherwise}$$

For this assumption, the complex model will be one with all possible interactions between the confounders up to the triple one:

Tidy Ouput for Model 2

R Code Start Over Run Code

```
triple_bin_log_model <- glm(chd69 ~ dibpat + age_bins * bmi_bins * smoke, fami
tidy(triple_bin_log_model) %>% mutate_if(is.numeric, round, 3)
```

```
# A tibble: 33 × 5
   term                              estimate std.error statistic p.value

 1 (Intercept)                         -4.45      0.719    -6.18    0
 2 dibpatType A                         0.732     0.144     5.07    0
 3 age_bins(42,45]                    -13.4     441.      -0.03    0.976
 4 age_bins(45,50]                      1.32      0.882     1.49    0.136
 5 age_bins(50,59]                      0.81      0.927     0.874   0.382
 6 bmi_bins(23,24.4]                    0.517     0.876     0.59    0.555
 7 bmi_bins(24.4,25.8]                  0.988     0.828     1.19    0.233
 8 bmi_bins(25.8,38.9]                  1.02      0.828     1.23    0.218
 9 smokeYes                             1.21      0.802     1.51    0.132
10 age_bins(42,45]:bmi_bins(23,24.4]   -0.576   659.       -0.001   0.999
# ℹ 23 more rows
```

Note

The estimate for `dibpatType A` is similar compared to `bin_log_model` (both estimates and their standard errors are practically equal!):

R Code Start Over Run Code

```
tidy(bin_log_model) %>% mutate_if(is.numeric, round, 3)
```

```
# A tibble: 9 × 5
  term                estimate std.error statistic p.value

1 (Intercept)           -4.24      0.246   -17.2     0
2 dibpatType A           0.73      0.144     5.08    0
3 age_bins(42,45]        0.19      0.23      0.826   0.409
4 age_bins(45,50]        0.687     0.197     3.48    0
5 age_bins(50,59]        1.02      0.187     5.47    0
6 bmi_bins(23,24.4]      0.413     0.208     1.98    0.048
7 bmi_bins(24.4,25.8]    0.569     0.205     2.78    0.005
8 bmi_bins(25.8,38.9]    0.796     0.199     4.01    0
9 smokeYes               0.704     0.139     5.05    0
```

Run Likelihood Ratio Test (LRT)

Then, we use the function `anova()` with the parameter `test = "LRT"`.

R Code Start Over Run Code

```
anova(bin_log_model, triple_bin_log_model, test = "LRT")  %>% mutate_if(is.num
p_values <- c(anova(bin_log_model, triple_bin_log_model, test = "LRT")$`Pr(>Ch
```

```
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      3092     1628.8
2      3068     1599.8 24   28.944    0.222
```

Note that we will be storing different p-values in vector `p_values`. Given the p-value above for the LRT, our simple model suffices.

4.4.2. Stress tests of the SECOND assumption (no interactions between X and the confounders)

For the second assumption, we will fit three other models.
Each of them will include the terms of our simple model `bin_log_model` plus one double interaction between X and a given confounder.
Then, we run the corresponding LRT via `anova()`.

Given confounder: Age

R Code Start Over Run Code

```
int_bin_log_model <-  glm(chd69 ~ dibpat + smoke + age_bins + bmi_bins + dibpa
anova(bin_log_model, int_bin_log_model, test = "LRT")  %>% mutate_if(is.numeri
p_values <- c(p_values, anova(bin_log_model, int_bin_log_model, test = "Chi")$
```

```
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      3092     1628.8
2      3089     1627.9  3    0.852    0.837
```

Given confounder: Smoke

R Code Start Over Run Code

```
int_bin_log_model_2 <-  glm(chd69 ~ dibpat + smoke + age_bins + bmi_bins + dib
anova(bin_log_model, int_bin_log_model_2, test = "LRT") %>% mutate_if(is.numer
p_values <- c(p_values, anova(bin_log_model, int_bin_log_model_2, test = "LRT"
```

```
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      3092     1628.8
2      3091     1627.7  1    1.133    0.287
```

Given confounder: BMI

R Code Start Over Run Code

```
int_bin_log_model_3 <-  glm(chd69 ~ dibpat + smoke + age_bins + bmi_bins + dib
anova(bin_log_model, int_bin_log_model_3, test = "LRT") %>% mutate_if(is.numer
p_values <- c(p_values, anova(bin_log_model, int_bin_log_model_3, test = "LRT"
```

```
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1      3092     1628.8
2      3089     1627.1  3     1.67    0.644
```

Conclusion From Three Models

- Given the p-values below for the LRT, our simple model suffices.

4.4.3. Stress test of the THIRD assumption (no additional confounders are necessary)

- Suppose we want to use the subject's cholesterol levels (`chol`) as an additional confounder besides the three ones already included in the simple `bin_log_model`.
- We will create another quartile strata for `chol` and fit another model called `chol_bin_log_model`.
- Then we run the corresponding LRT.

Make quartile strata for `chol`

R Code Start Over Run Code

```
training_data <- training_data %>%
  mutate(chol_bins = cut(chol, breaks = c(min(chol), quantile(chol, (1:3) / 4)
levels(training_data$chol_bins)
```

```
[1] "[103,197]" "(197,223]" "(223,253]" "(253,645]"
```

Fir Model with `chol` and Run LRT

R Code Start Over Run Code

```
attach(training_data)
chol_bin_log_model <- glm(chd69 ~ dibpat + age_bins + smoke + bmi_bins + chol_
```

```
The following objects are masked from training_data (pos = 3):

    age, age_bins, bmi, bmi_bins, chd69, chol, dibpat, smoke


The following objects are masked from training_data (pos = 4):

    age, age_bins, bmi, chd69, chol, dibpat, smoke
```

R Code Start Over Run Code

```r
anova(bin_log_model, chol_bin_log_model, test = "LRT") %>% mutate_if(is.numeri
p_values <- c(p_values, anova(bin_log_model, chol_bin_log_model, test = "LRT")
```

```
  Resid. Df Resid. Dev Df Deviance  Pr(>Chi)
1      3092     1628.8
2      3089     1580.3  3   48.523 < 2.2e-16 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Conclusion

- Note that in this case, we reject H_0 in our LRT.
- Hence we need to consider `chol` in our logistic regression model!
- Therefore, `bin_log_model` DOES NOT fulfill this assumption.

But …

Testing the third assumption is challenging when assessing causation in observational studies involving many different confounders, such as the WCGS case.

Hence, we can think about a practical solution: automating the model building process and selection (confounder by confounder) while testing the three assumptions.

> Heads-up: The above idea is a crucial part of your work as a Data Scientist when doing model building and selection in observational studies to detect causation. Moreover, this process involves multiple testing! This is the reason we stored all the previous `p_values`.

Adjust for Multiple Testing!

R Code Start Over Run Code

```
adj_p_values <- matrix(data = p.adjust(p_values, method = "bonferroni"), nrow
adj_p_values
```

```
     [,1] [,2] [,3] [,4]         [,5]
[1,]    1    1    1    1 8.239119e-10
```

Conclusion

The last adjusted p-value from left to right involves the LRT of `bin_log_model` versus `chol_bin_log_model`.
We would need to proceed with `chol_bin_log_model` and restart the model checking since the first assumption.

> Testing the third assumption will involve an additional confounder from `wcgs`.

5. Wrapping Up

We have the following parting thoughts for today:

- We have made a connection between stratification and regression.
- In some contexts, we build regression models with prediction as the main goal.
- In others (like today!), we build regression models with explanation as the main goal.

Wrapping Up (Continued)

- For causal interpretations, various assumptions must be met. These assumptions are pretty strong!
- Some of them are testable/checkable (typically having to do with appropriate model specification issues like smoothness, lack of interactions between the variable of interest and confounders, etc.).
- We almost always have at least one untestable assumption, e.g., no unobserved confounders.

DSCI 554 - Experimentation and Causal Inference

Speaker notes